11.8 Tilted Equilibria

WORKSHEET

The question

In this question, $n \geq 2$.

(i) A solid, of uniform density, is formed by rotating through 360° about the y-axis the region bounded by the part of the curve $r^{n-1}y = r^n - x^n$ with $0 \leq x \leq r$, and the x- and y-axes. Show that the y-coordinate of the centre of mass of this solid is $\dfrac{nr}{2(n+1)}$.

(ii) Show that the normal to the curve $r^{n-1}y = r^n - x^n$ at the point $\left(rp, r(1-p^n)\right)$, where $0 < p \leq 1$, meets the y-axis at $(0, Y)$, where $Y = r\left(1 - p^n - \dfrac{1}{np^{n-2}}\right)$.

In the case $n = 4$, show that the greatest value of Y is $\frac{1}{4}r$.

(iii) A solid is formed by rotating through 360° about the y-axis the region bounded by the curves $r^3y = r^4 - x^4$ and $ry = -(r^2 - x^2)$, both for $0 \leq x \leq r$. A and B are the points $(0, -r)$ and $(0, r)$, respectively, on the surface of the solid.

Show that the solid can rest in equilibrium on a horizontal surface with the vector $\overrightarrow{AB}$ at three different, non-zero, angles to the upward vertical. You should not attempt to find these angles.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself A solid rests on a table, touching at a single point X of its curved surface. The forces on it are its weight (acting at the centre of mass G) and the reaction at X. When are these in equilibrium?

STAGE 1 The centre of mass

Ask yourself Slice the solid into thin cylindrical shells of radius x, thickness δx and height y. What is the volume of a shell, and where is its centre of mass?

STAGE 2 Where normals meet the axis

Ask yourself For $n = 4$, $Y = r\left(1 - p^4 - \frac{1}{4p^2}\right)$. Where is it greatest? What happens as $p \to 0$, and at $p = 1$?

STAGE 3 The spinning top

Ask yourself The solid in part (iii) is a dome ($n = 4$) on top of an upside-down paraboloid ($n = 2$). Where is its centre of mass?

STAGE 4 Counting equilibria

Ask yourself A tilted equilibrium needs a point (not A or B) whose normal passes through G. On each part of the surface, how many points have normals meeting the axis at height $\frac{3r}{35}$?

Try this next

1 **A self-righting toy.** (From STEP II, 1996.) A toy is made of a solid cone of height λa fixed to a solid hemisphere of radius a along their common circular face. Show that if $\lambda < \sqrt{3}$ the toy returns upright when placed with its hemisphere on a table, but that if $\lambda > \sqrt{3}$ it may overbalance. (*Hint:* where must G be relative to the centre of the hemisphere?) Show that if it is placed with the cone touching the table, it stays there whatever λ.

2 **A sphere.** For a uniform solid sphere, every normal passes through the centre. What does that say about its equilibria? What if the sphere's centre of mass is displaced slightly from its centre?

3 **An ellipse.** Show that an elliptical disc rolling in its own plane on a line has exactly four equilibria, and identify which are stable.

4 **Two paraboloids.** Replace the dome in part (iii) by a paraboloid $ry = r^2 - x^2$, so that the solid is symmetric about the x-axis. Show that G is at the origin, find the tilted equilibria, and explain why they come in pairs with equal tilts.

Further reading

- **Péter Várkonyi & Gábor Domokos, "Static equilibria of rigid bodies: dice, pebbles, and the Poincaré–Hopf theorem"**, *Journal of Nonlinear Science* 16 (2006). The Gömböc paper.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), on equilibrium and stability.
- **Chris Rorres, "Completing Book II of Archimedes's *On Floating Bodies*"**, *Mathematical Intelligencer* 26 (2004). Archimedes' stability analysis, with modern pictures.

The full guided solution, and what the question leads to at university, is in Section 11.8:
https://beyondstep.keithmcnulty.org/ch11/tilted-equilibria.html